Modules – Config

Menu: Modules > Config

Configures the hardware settings of the connected module.

Module Identity

The underlined line at the top shows which module the module pages are editing — the module number, the board's description (or reported label), and its connection state. It updates live while the page is open. If it turns red with **"Two boards are answering as module N!"**, two powered boards share the same module ID — disconnect one and give it its own ID before continuing.

Below it, while the module is connected, is the board type it reports, e.g. **Teensy Rate** or **ESP Rate**. Nothing is shown while it is not connected.

When a new board is connected, a further line reads **New board: *type*. Set a New ID, then Save.** New and blank boards start at ID 25 and only report their status until they are given an ID from 0 to 7. This line is for boards on Ethernet or CAN. An ESP32 on WiFi is set up from **Modules > Communication > WiFi**, which gives it its ID at the same time.

Setting	Description
Module to edit	The module to edit, listed as *N (Connected)* or *N (Not connected)*. All module pages (Config, Communication, Pins, Relay Pins, Valves) and the Send button work on this module; other connected modules ignore the settings. A new choice takes effect straight away, without Save. It is refused while any module page has unsaved changes.
New Module ID	Leave blank unless changing the board's module ID. See **Changing a Module ID** below.
New Board Description	Leave blank to keep the current description. Up to 16 characters (e.g. "Left Tank"). Press Save, then Send. The description is sent once and stored in the module's own memory, where it is kept through firmware updates. The box is blank again afterwards.

Changing a Module ID

An ID change is a step of its own and is sent as soon as it is confirmed, without pressing Send.

1. Send any saved changes first. While changes on any module page are unsaved, or saved but not yet sent, the New Module ID box can't be used.
2. Connect **only the board to be changed**. Every board that hears the new ID takes it.
3. Enter the new ID and press Save. The rest of the page is locked while a new ID is entered.
4. Confirm with **OK**. The board's sensor pins go with the ID. The board restarts and reconnects as the new module number, keeping its settings, and all module pages switch to the new ID.

The change is refused, and the module ID stays as it was, when no board is connected, when more than one board is connected, or when two boards are answering as the same module. A message says which. Cancel in the confirmation box also keeps the old ID.

> **Note:** A new board (ID 25) can't be sent its sensor pins until it has an ID. After it reconnects under its new ID, the Send button stays highlighted. Press Send to give it its pins.

> **Note:** A board with no description reports its built-in hardware (MAC) number instead, e.g. **MAC 04E9E5A1B2C3** — unique to every board. This is how RateController tells new boards apart and detects duplicate module IDs. Give each board a friendly description once it is set up. (Nano boards do not report a label.)

1. Relay Type

Value	Description
0	No relays
1	GPIOs — use micro-controller pins directly
2	PCA9555 — 8-relay module
3	PCA9555 — 16-relay module
4	MCP23017
5	PCA9685
6	PCF8574

2. Pin Assignments by Board

Parameter	RC5 (Nano)	RC8 (Nano)	RC11 (Teensy)	RC11-2 (Teensy)	RC12 (Nano)	RC12-3 (Nano)	RC15 (ESP32)
Sensor Count	2	2	2	2	1	2	2
Relay Control	4	4	1	1	2	4	5
Flow 1	2	2	28	28	D3	3	17
Flow 2	3	3	29	29	—	2	16
DIR 1	4	4	37	37	D6	4	32
DIR 2	6	6	14	14	—	6	25
PWM 1	5	5	36	36	D9	5	33
PWM 2	9	9	15	15	—	9	26
Work Pin	—	—	—	30	—	15	—
Pressure Pin	—	—	—	40	—	14	—

3. Nano Pin Reference

Pin Name	Number
D0–D13	0–13
A0–A7	14–21

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

Modules – Pins

Menu: Modules > Pins

Manually assigns the GPIO pins used for sensors and motor/valve control on the module. The grid shows one row per sensor (the sensor count is set in Modules > Config). Pin values range from 0–50. A blank value (displayed as "–") means the pin is not assigned.

Grid Column	Description
Sensor	Sensor/product number (read-only)
Flow	Input pin for the flow sensor
Dir	Output pin for the motor/valve direction channel
PWM	PWM output pin for the motor/valve
Bin	Input pin for the bin-empty sensor. Leave unassigned ("–") if the product has no bin sensor — the bin alarm is off for that product.
Invert	Inverts the bin sensor reading. Unchecked: an open input reads *empty* (sensor closes to ground while material is present). Checked: a grounded input reads *empty*.

Setting	Description
Work Pin	Input pin for the work/implement switch
Momentary	Enables momentary switch behavior on the work pin
Invert Work Switch	Inverts the work switch reading to suit the sensor type. Unchecked: an open input reads as *working* (normally-closed sensor). Checked: a grounded input reads as *working* (normally-open sensor). In momentary mode the toggle occurs on the opposite edge. Older module firmware ignores this setting.
Pressure Pin	Analog input pin for the pressure sensor

Press **Rescan** to clear all pin assignments. Duplicate pin numbers are rejected on Save.

> **Tip:** When a board type is selected in Modules > Config, pressing **Defaults** populates these pin values automatically. Only change them if your hardware differs from the standard layout.

Bin-Empty Alarm

When a Bin pin is assigned, the module watches that input and reports the bin state to RateController. The bin sensor input uses an internal pull-up, so a simple switch or level sensor wired between the pin and ground is all that is required. The module filters the reading — the sensor must hold its state for about 2.5 seconds before the reported state changes, so brief flickers from material sloshing are ignored.

- The alarm is only raised **while the master switch is on** — an empty bin at start-up does not alarm until you begin applying.
- When it triggers, the alarm button shows **Bin Empty** with the product name, the product's tile turns red, and on the selected product the quantity caption changes to **Bin Empty** with the quantity value shown as a red 0.
- The displayed 0 is display-only — the recorded tank amount and quantity applied are not changed, and the real value returns as soon as the alarm clears.
- The alarm clears immediately when the bin refills or the master switch is turned off. Tap the alarm button to silence the buzzer.

> **Tip:** Use a plain digital pin for the bin sensor. Inputs with analog conditioning on the board (such as the pressure input) load the internal pull-up and may not read reliably as a digital input.

Controls

Control	Action
⊗ Cancel	Discard changes
Save	Save settings

Modules – Relay Pins

Menu: Modules > Relay Pins

Assigns the physical GPIO pins to each of the 16 relay outputs on the module. Pin values range from 0–50. A value of 255 (displayed as "–") means the relay pin is not assigned.

- **Low Relays (1–8):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.
- **High Relays (9–16):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.

Press **Rescan** to clear all relay pin assignments.

> **Tip:** When a board type is selected and **Defaults** is pressed in Modules > Config, relay pins are populated automatically for the chosen hardware.

Controls

Control	Action
⊗ Cancel	Discard changes
Save	Save settings

Modules – Valves

Menu: Modules > Valves

Selects the wiring mode for the valve or motor driver.

Option	Description
2-Wire	Motorized or latching valve — the driver actively powers the valve in both directions to open and close it. 
3-Wire	One switched signal wire, one power wire and one ground wire. Compatible with standard solenoid valves and motorized ball valves that manage open/close internally. Default for most installations. 

Select the mode that matches how your valve or motor driver is wired to the module.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings